

An Analytical Framework for Error and Hallucination in Deployed Generative AI Systems

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Abstract

Generative AI applications increasingly combine a trained model with additional material: designated source documents, retrieved evidence, tool outputs, prior interactions, and technical, procedural, or organizational controls. Because of this, a flaw in the response ultimately delivered to the user may not originate in the trained model—it may originate in how that additional material was gathered, assembled, or filtered. Moreover, the information that should authoritatively govern a given claim—its governing reference—can differ from one task to the next. This paper develops an analytical framework to anatomize the response error, based on three distinctions: the information available for response generation is not the same as the information authoritative for its evaluation; a discrepancy between a claim and its governing reference is a separate question from why that discrepancy occurred; and whether a discrepancy occurred is separate from how much it matters. The framework treats hallucination and task-obligation fidelity error as related but potentially overlapping problems. Hallucination is a generated representational assertion that is contradicted by, or lacks support required from, its governing reference. A task-obligation fidelity error is a failure to meet a requirement set by a governing source, instruction, contextual record, or authorized project state. A behavioral failure to meet a requirement need not be a hallucination, while a false assertion about an instruction, source, contextual record, or authorized project state can exhibit both errors at once. The framework is extended to consider temporal change, project state in continuing workflows, retrieval-augmented generation (RAG), the possibility that errors propagate forward, and system-level controls. The result is an authority-first evaluation sequence: identify what governs a claim, and what evidence is needed, before classifying a discrepancy, tracing its cause, or deciding on a governance response.

Key words: AI governance; AI evaluation; model error; hallucination; factuality; faithfulness; governing reference; task-obligation fidelity; project-state fidelity; RAG; error propagation.

A. Introduction

Generative AI applications have become increasingly widespread. They are used in research, customer service, software development, financial analysis, compliance, education, government, and many other activities. The applications have also changed. Many early uses involved a single, stand-alone question or a request to generate or edit a piece of content. Applications today are more mature: they combine an AI model with an organization's own information, software tools, operating processes, and human decision-making. Many of these applications also continue across time rather than ending after a single response.

This evolution creates two related problems for AI governance and evaluation. First, an appropriate response may depend on information not contained in the current input. Beyond the model's trained knowledge, a response may depend on designated source material, retrieved evidence, tool results, prior exchanges in the conversation, and decisions made earlier in an ongoing project. These additional sources can be incomplete or inconsistent with one another, and can differ in their authority for a given task. A system must determine not only what information is available and relevant, but also which information should actually govern the response. Before an evaluator can decide whether an error occurred, the governing reference that applies must first be identified.

Second, the consequences of an erroneous response may become more serious. An incorrect statement in a casual conversation may do little harm. The same statement, if it appears in a research record, a compliance analysis, or an organizational decision process, may be relied upon or passed to another person or system. In a continuing project, it may also become part of the context feeding a later AI response. An error therefore can outlive the exchange in which it was made and—if it is retained and reused — may contribute to errors in later responses.

These problems are often discussed under the single term hallucination, but the term is used inconsistently. Some definitions emphasize generated content that is factually false; others include content that is simply unsupported by available evidence; still others emphasize content that is inconsistent with a supplied document or source, whether or not that content happens to be true ([1]–[5]). Under this last definition, a statement can be factually accurate and still inappropriate for a source-based task because it changes or contradicts the source it was supposed to represent faithfully. Separately, a discrepancy—that is, a failure of generated content to meet the applicable evaluative standard—can originate in the trained model, in retrieval, in how context was assembled, in generation itself, or in a control surrounding the model. Treating all of these different relations and causes as one undifferentiated failure mode obscures both evaluation and remedy.

Existing work has already drawn some useful distinctions: between claims that contradict a supplied source and claims that cannot be verified from it (often called intrinsic versus extrinsic hallucination), and between conformity with supplied information and agreement with external, real-world facts (often described as faithfulness versus factuality) ([6], [7]). This paper begins one step earlier, by asking which reference should have governed a particular claim in the first place, at the time relevant to the task. Rather than proposing a new taxonomy of hallucination, it offers an authority-first evaluation sequence connecting reference selection, discrepancy classification, causal investigation, consequence, and governance response. Before asking why an AI system produced an error, an evaluator should first identify the governing reference and determine what discrepancy—if any—occurred. Causal attribution and consequence assessment should come afterwards. Under this sequence, a retrieval failure is not itself a hallucination if the system abstains from answering; conversely, a fabricated claim in the delivered response is still a hallucination even if a retrieval failure contributed to producing it.

The framework also distinguishes discrepancies in the content of a claim from failures to meet the obligations of a task. It classifies as hallucination a generated representational assertion about an external fact, a designated source, an instruction, a contextual record, or an authorized project state when that assertion is contradicted by, or lacks support required from, its governing reference. Here, authorized project state means the set of accepted decisions, requirements, and statuses governing ongoing work at the time relevant to the task. We use the term task-obligation fidelity for conformity with requirements set by a governing source, instruction, contextual record, or authorized project state. A task-obligation fidelity error is a failure to meet one of those requirements. Hallucination and task-obligation fidelity error can overlap.

Our framework is analytical rather than empirical. It makes explicit the task, governing reference, observed discrepancy, supported causal attribution, consequence, and governance response, while allowing the governing reference or causal attribution to remain unresolved when the evidence does not support a firm conclusion. The framework thereby offers a structured tool for diagnosing errors in delivered AI responses and for considering corresponding governance actions.

The paper develops the framework in three stages. Sections B–E build the core, single-interaction analysis: how a system produces a response, how a governing reference is selected, how a claim is

classified, and how a cause is attributed. Sections F and G extend that analysis across interactions over time and into the internal workings of context construction, respectively. Section H turns to consequences, controls, and application, and Section I discusses the framework’s limitations and concludes.

B. A System-Level Production Model

A deployed generative AI system produces a response through several analytically distinct stages. At interaction time t , the system receives the current input x_t , has access to the prior interaction history $\mathcal{H}_{t-1}$, and may draw on external information resources E_t , such as documents, databases, tool-accessible information, or current observations.

Guided by the input, a context-construction operator R^L may retrieve material from E_t and then select, compress, order, and label relevant information from both the retrieved material and $\mathcal{H}_{t-1}$. The superscript L denotes the effective context constraint: the practical limit on the amount and form of information that can be supplied to the model for this interaction. Subject to that constraint, the operator produces the operative context:

$$C_t = R^L(x_t, \mathcal{H}_{t-1}, E_t). \quad (1)$$

The operative context is therefore a constructed subset of the information nominally available to the system. Material may exist in a file, database, or prior exchange yet remain outside C_t because it was not retrieved, did not fit within the constraint, was displaced by other material, or was compressed in a way that altered its provenance or authority. Inclusion in the operative context does not guarantee effective use: model performance can depend on where relevant information appears within a long context ([8]).

Let $M(t_0)$ denote the trained model, where t_0 is its training cutoff, and let P_M denote the response distribution produced by the deployed generation process using that model. Conditional on the current input and operative context, the model produces a candidate response drawn from this distribution:

$$\tilde{y}_t \sim P_M(\cdot | x_t, C_t). \quad (2)$$

The interval between t_0 and t creates the possibility that the model’s trained knowledge is outdated, although the interval alone does not establish an error. The current input influences generation both directly, by specifying the task, and indirectly, by shaping the construction of C_t .

Before the candidate response reaches the user, a post-generation intervention layer may transform, qualify, redact, refuse, or block it. Let Σ denote the rules and controls in that layer, and I_Σ the operator applying them. The delivered response is:

$$y_t = I_\Sigma(\tilde{y}_t, x_t, C_t). \quad (3)$$

The delivered response may therefore differ from the candidate response because of compliance, safety, confidentiality, privacy, security, or output-validation requirements. Such intervention may prevent harmful content from reaching the user, but it may also remove qualifications, alter meaning, or introduce new discrepancies. Distinguishing the candidate response from the delivered response is essential for diagnosis.

Equations (1)–(3) identify four analytically distinct influences on the delivered response: the current input, context construction, model generation, and post-generation intervention. The delivered response is thus shaped not only by the trained model but also by the system’s deployment configuration. Evaluation requires determining which part of the response is under assessment and which reference governs that assessment.

C. Governing References and Assertion-Reference Discrepancy

Evaluation of the delivered response begins by identifying the claims within the response that can be assessed against a governing reference. Following the claim-level approach used in factuality evaluation [9], we call each such claim an assertion. An assertion is an explicit or reasonably implied claim conveyed by the response whose relation to a governing reference can be assessed. A response may contain several assertions when it conveys independently assessable claims, and those assertions may be governed by different references.

Once assertions have been identified, each must be evaluated against the reference that governs it for the task. Selecting that reference requires both a task specification q_t , which states what the system is being asked to do, and an authority policy Π_t , which determines how authority is assigned when multiple references may appear relevant. Establishing the authority policy in advance limits retrospective selection of a reference that favors a preferred outcome.

Let a_{tj} denote assertion j in the delivered response, and let k_{tj} denote its role: external factual claim, designated-source claim, claim about an applicable instruction, contextual-record claim, or project-state claim. Let ρ denote the reference-selection rule. The governing reference for assertion a_{tj} is:

$$r_{tj}^* = \rho(q_t, k_{tj}; \Pi_t). \quad (4)$$

Depending on the assertion’s role, the governing reference may be current external information, a designated source and its provenance, an applicable instruction, an authoritative contextual record, or the authorized state of a continuing project. When no governing reference can be established, authority remains unresolved.

The governing reference is an evaluative standard; it is not necessarily information the model received. In particular, the operative context C_t supplied to the model must be distinguished from any contextual record designated as authoritative for evaluation. A response may be consistent with its operative context yet conflict with its governing reference.

Table 1 summarizes the assertion roles used in Equation (4), their ordinary governing references, and representative illustrations.

Table 1. Assertion Roles and Governing References

Assertion role	Ordinary governing reference	Illustration
External factual claim	Current verifiable information	Who currently holds an office; a current price or rule.
Designated-source claim	Designated source and its provenance	What a supplied paper states, including its qualifications.
Claim about an applicable instruction	Applicable instruction	Whether the instruction required a specified scope, format, or prohibited transformation.
Contextual claim	Authoritative contextual record	What was established earlier in the operative exchange.
Project-state statement	Authorized project state	Whether a proposal was accepted, revised, or superseded.

Once the governing reference has been selected, assertion a_{tj} can be compared with it. Let v denote the comparison rule. The assertion–reference relation is:

$$V_{tj} = v(a_{tj}, r_{tj}^*), V_{tj} \in \{\text{supported}, \text{contradicted}, \text{unsupported}, \text{unresolved}\}. \quad (5)$$

A relation is *supported* when the governing reference provides the basis required by the task; *contradicted* when the reference conflicts with the assertion; *unsupported* when support required by the task is absent; and *unresolved* when the evaluator cannot determine the relation.

For discrepancy reporting, Equation (6) maps these relations into a binary outcome:

$$D_{tj} = \begin{cases} 1, & V_{tj} \in \{\text{contradicted}, \text{unsupported}\}, \\ 0, & V_{tj} = \text{supported}, \\ \text{unresolved}, & V_{tj} = \text{unresolved}. \end{cases} \quad (6)$$

Equation (6) applies only when the delivered response presents an assertion–reference question. Within that scope, a value of one denotes a discrepancy relative to the governing reference, while zero denotes an established absence of discrepancy. An unresolved outcome does not provide evidence of successful performance and should not be treated as zero. Response that does not present an assertion–reference question falls outside Equations (5) and (6) and should be reported separately as not applicable. “Not applicable” is thus a scope determination, not an assertion–reference relation.

D. Hallucination and Task-Obligation Fidelity Errors

The framework distinguishes two classifications of discrepancy: hallucination and task-obligation fidelity error. They answer different evaluative questions and may occur separately or together. Hallucination concerns the representational content of a generated assertion. Task-obligation fidelity concerns whether the delivered response conforms to an applicable requirement.

A representational assertion is an assertion about an external fact, designated source, applicable instruction, authoritative contextual record, or authorized project state. Such an assertion is classified as hallucination when it is contradicted by, or lacks support required from, its governing reference. The classification therefore concerns the relationship between the representational content of the delivered response and the authoritative reference selected under Section C.

Task-obligation fidelity means conformity with requirements established by a governing source, instruction, contextual record, or authorized project state. A task-obligation fidelity error occurs when the delivered response departs from one of those requirements. The requirement may be behavioral or structural, such as a word limit, or it may require the response to represent a governing source faithfully.

The two classifications may occur separately or together. Consider a request for a source-faithful summary of no more than 200 words. If the system produces an accurate 300-word summary, the excess length is a task-obligation fidelity error: the response fails to meet a requirement of the instruction. It is not hallucination, however, because the excess length is a behavioral departure rather than a representational assertion. If the system additionally asserts that the 300-word summary complies with the instruction, that assertion is contradicted by the governing reference—the instruction itself—and is therefore hallucination. The latter response exhibits both classifications.

Allowing the discrepancy classifications to overlap avoids two distortions. Treating every task failure as hallucination would erase the distinction between a formatting violation and a fabricated citation.

Conversely, excluding project-state discrepancies from hallucination would place structurally similar false assertions—such as “the project adopted the proposal” and “the user approved the proposal”—into different categories solely because one concerns project state and the other concerns contextual record. The framework therefore permits overlap when a representational assertion about a governing source, instruction, contextual record, or authorized project state is contradicted by, or unsupported by, its governing reference.

E. Causal Attribution

Once an assertion–reference discrepancy has been identified and classified, the next step is causal attribution: determining which stage or stages of the system’s production process contributed to the discrepancy and what evidence supports that conclusion. A discrepancy may arise through different mechanisms ([10]). For example, a hallucination involving a designated source may result from failure to retrieve the source, omission of the relevant passage during context construction, or generation that disregards evidence already present in the operative context. A discrepancy may also reflect limitations in the trained model or effects introduced by post-generation intervention. The occurrence of a discrepancy alone does not distinguish among these mechanisms, and a control directed at one mechanism may have little effect on another.

For attribution reporting, the framework uses four diagnostic stage labels.

- **M** denotes a contribution attributable to deficient or outdated knowledge or capability in the trained model.
- **R** denotes a contribution from context construction, including retrieval, memory, compression, ordering, or authority labeling.
- **G** denotes a contribution from failure to use or reconcile information in the operative context during generation.
- **I** denotes a contribution from post-generation intervention.

These labels represent supported attribution findings; they are not alternative names for the production components introduced in Section B. In particular, **M** is distinct from the trained model $M(t_0)$, **R** from the context-construction operator R^L , and **I** from the intervention operator I_Σ . **G** identifies a generation-stage contribution rather than a separately defined operator.

Let Z_{tj} denote the supported attribution for assertion a_{tj} . It may identify a single contributing stage or a nonempty set of jointly contributing stages. When the available evidence supports none of these findings, attribution remains unresolved:

$$Z_{tj} \in \mathcal{P}(\{M, R, G, I\}) \setminus \{\emptyset\} \cup \{\text{unresolved}\}. \quad (7)$$

The power-set operator $\mathcal{P}$ permits attribution to one stage or to several jointly contributing stages; removing the empty set ensures that a resolved attribution identifies at least one contribution. The unresolved value records that the available evidence does not support a stage-level attribution.

Causal attribution remains analytically separate from both hallucination and task-obligation fidelity classification. A hallucination can be established even when Z_{tj} is unresolved, because the governing reference may show that the assertion is discrepant without revealing how the system produced it. Conversely, neither a retrieval failure nor the mere presence of a control-layer transformation establishes hallucination. Hallucination arises only when a representational assertion in the delivered response is contradicted by, or unsupported by, its governing reference.

Attribution requires evidence about the production stages that could plausibly have contributed to the delivered response. Logs may show what was retrieved and packed; a pre-intervention candidate response may reveal whether a later intervention changed the answer; repeated trials may indicate whether an error is systematic; and controlled interventions may help distinguish competing explanations. Even after examining such evidence, attribution may remain unresolved.

Table 2 illustrates the separation between discrepancy occurrence and supported causal attribution.

Table 2. Discrepancy Occurrence and Possible Causal Attribution

Observed condition	Discrepancy occurrence	Possible attribution
A retrieval system returns no relevant evidence and the model abstains.	Availability failure; no assertion discrepancy.	R
Retrieval fails and the model invents a source-backed answer.	Hallucination involving a designated source.	{R, G}
Correct evidence is present in the operative context, but the response contradicts it.	Hallucination involving a designated source.	G
A current factual question is answered from outdated model knowledge.	Factual hallucination if assertion contradicted by the current reference.	M; {M, G} if elaborated
A control inserts an unsupported explanation for a refusal.	Possible factual or contextual hallucination.	I

Table 2 shows that the mapping between discrepancy classification and supported causal attribution is not one-to-one. The same hallucination involving a designated source may be attributed to **R**, **G**, or both, depending on what evidence entered the operative context and how the generator used it. Similarly, an infrastructure failure may produce no hallucination when the system abstains. A hallucination or task-obligation fidelity error therefore identifies what occurred relative to the governing reference; it does not by itself establish how the deployed system produced the discrepancy.

F. Time, Project State, and Error Propagation

Evaluation occurs at a specific interaction time and within a specific project state. Both dimensions may affect governing references, discrepancy classification, and causal attribution. For external facts, the governing time is the time specified or implied by the task, which may differ from the interaction or evaluation time. For continuing work, project state determines which contextual records, instructions, and authorized decisions govern the task. Evaluation must therefore be anchored to the time and project state relevant to the delivered response.

Let t denote the interaction time and s_t the authorized project state at that time. The governing reference for an assertion may depend on both, but the task may designate a different reference time. For an external-factual claim, the governing reference is the factual state at the time relevant to the task. For a designated-source claim, it is the designated source with its applicable provenance and version. For claims about instructions, contextual records, or project state, it is the authoritative version of those materials under the applicable project state. A discrepancy may therefore arise because the delivered response reflects the wrong factual time, an outdated instruction, or a superseded project decision.

Project state may also create conditions under which errors persist or contribute to later errors. A representational assertion that conflicts with the governing reference is hallucination even if it was accurate at another time, provided that the task requires the later state. A task-obligation fidelity error may

likewise arise when an instruction or authorized decision has changed. Classification follows the governing reference applicable to the task, not the state the system may have implicitly assumed.

Possible propagation arises when an earlier discrepant assertion is retained or reused and thereby contributes to a later discrepancy. A hallucination about project state may lead to later task-obligation fidelity errors if the system acts on the incorrect assertion. Persistence alone does not establish propagation: the assertion may never be retrieved again, may be corrected, or may be irrelevant to the later task. Long-term memory design can affect what information is retained and recovered across interactions, although successful recall does not by itself preserve the authority or status of the recovered information [11].

Establishing propagation requires evidence that the earlier assertion was retained or reused and that this reuse affected the later response. Mere temporal sequence or textual similarity is insufficient. A later discrepancy retains its own occurrence classification and may have a different supported attribution under Equation (7).

Time and project state therefore serve two roles. First, they help determine the governing references used for discrepancy classification. Second, they identify conditions under which an erroneous assertion may persist and affect later work. Evaluation should record the interaction time, the time relevant to the task, and the applicable project state whenever those distinctions could change the result.

G. Retrieval-Augmented Generation

Where Section F extends the framework across interactions, Section G examines one component of the production process in greater detail: the construction of the operative context C_t from external information resources E_t . Retrieval-augmented generation combines model generation with retrieved external information for knowledge-intensive tasks [12]. In the present framework, the context-construction operator R^L introduced in Section B is decomposed into query construction, retrieval and ranking, and context packing. This decomposition clarifies how external information enters the operative context and how failures in those steps may contribute to discrepancies in the delivered response.

Let $K_t \subseteq E_t$ denote the corpus available to retrieval at interaction time t . Let Q denote a query constructor, Ret_ϕ a retrieval and ranking operator configured by retrieval parameters ϕ , and Pack^L a packing function subject to the effective context constraint L . To distinguish retrieval queries from the task specification q_t introduced in Section C, let q_t^R denote the retrieval query. The RAG pipeline can be represented as:

$$q_t^R = Q(x_t, \mathcal{H}_{t-1}), J_t = \text{Ret}_\phi(q_t^R, K_t), C_t^R = \text{Pack}^L(x_t, \mathcal{H}_{t-1}, J_t). \quad (8)$$

In a simple RAG deployment, C_t^R may constitute the operative context C_t in Equation (1). In a broader system, C_t^R may be only one component of C_t , alongside tool results, designated sources, or other records. Equation (8) identifies three distinct failure points: query construction may misstate the information need, retrieval may fail to return relevant material, and packing may omit or distort content, provenance, or authority. Any of these is a context-construction failure attributable to R under Section E, but none is by itself hallucination. These stages correspond to documented engineering and taxonomic treatments of RAG failure [13], [14].

RAG does not determine governing references. The governing reference for an assertion a_{ij} is selected under Section C and may be current external information, a designated source, an applicable instruction, an authoritative contextual record, or authorized project state. Retrieved material may or may not contain that reference. Hallucination is established only when a representational assertion is contradicted by, or unsupported by, its governing reference, regardless of whether the relevant passage was retrieved into J_t .

or included in C_t^R or C_t . Retrieval failures may affect whether a discrepancy occurs and how it is attributed, but they do not themselves determine the classification.

RAG may reduce discrepancies when retrieval returns relevant evidence and the generator uses it. It may increase discrepancies when retrieval returns irrelevant or misleading evidence, when packing displaces or compresses relevant material, or when the generator disregards evidence present in C_t . Because the operative context is constructed subject to constraint L , relevant material may be omitted even when retrieval succeeds. RAG therefore introduces both opportunities and risks: it can improve grounding when retrieval and packing succeed, but it can also create or exacerbate discrepancies when they fail.

The authority-first evaluation sequence ensures that RAG is analyzed correctly. The evaluator first identifies the governing reference for each assertion, then determines whether a discrepancy occurred, and only then considers whether retrieval, packing, or generation contributed to that discrepancy. Retrieval failures do not alter the governing reference and do not establish hallucination. Conversely, a fabricated claim in the delivered response is hallucination even when a retrieval failure contributed to producing it. RAG thus affects the likelihood and attribution of discrepancies but does not alter the evaluative standard.

H. Consequence, Controls, and Application

Once a discrepancy has been classified and causal attribution has been investigated as far as the evidence permits, the evaluator may consider consequences and governance responses. Consequence assessment concerns the effect of a discrepancy on the task, the project, or the organization. Governance response concerns actions taken to mitigate harm, prevent recurrence, or improve system performance. These steps follow classification and causal investigation; they do not determine them.

Consequences depend on both the nature of the discrepancy and the context in which it occurs. A hallucination in a casual conversation may have little effect, while the same hallucination in a compliance analysis or research record may be relied upon or propagated forward. A task-obligation fidelity error may be inconsequential when the requirement is cosmetic, but serious when the requirement concerns scope, confidentiality, or legal compliance. In a continuing project, an erroneous assertion may enter the operative context of later responses and contribute to further discrepancies.

Governance responses depend on both discrepancy classification and supported causal attribution. When a discrepancy is attributable to M , model-level interventions such as updating, fine-tuning, or replacing the model may be appropriate. When the deficiency concerns the task specification or supplied context, prompt or context-design changes may instead be warranted. When attribution includes R , improvements to retrieval, ranking, memory, or packing may be appropriate. When it includes G , controls supporting the use or reconciliation of operative-context information may be needed. When it includes I , post-generation controls may require revision. When attribution is unresolved, monitoring, logging, or controlled experimentation may be needed to gather additional evidence.

Authority-first evaluation helps prevent misdiagnosis. A retrieval failure is not itself hallucination if the system abstains; a delivered hallucination remains hallucination even when retrieval failed. A formatting violation is not hallucination; a false assertion about an instruction may be both hallucination and a task-obligation fidelity error. Governance responses should therefore be tied to the discrepancy classification and supported attribution, not to assumptions about where errors usually arise.

The framework applies across a wide range of deployments. In source-based tasks, it clarifies whether discrepancies arise from misreading the source, mispacking the source, or disregarding the source. In factual tasks, it distinguishes outdated model knowledge from retrieval failures. In continuing projects, it identifies how discrepancies propagate and how project-state fidelity can be maintained. In safety-critical

deployments, it helps separate model-level risks from system-level risks introduced by retrieval, packing, or intervention.

I. Discussion and Conclusion

The framework is analytical rather than empirical. It does not attempt to measure the frequency of hallucination, the prevalence of task-obligation fidelity errors, or the performance of specific retrieval or packing algorithms. It identifies the inputs to governing-reference selection but does not supply a substantive authority policy for every domain. When authority cannot be established, unresolved authority is an explicit outcome. The framework likewise does not force causal attribution when evidence is insufficient, or quantify consequences and prescribe governance responses independently of organizational context.

The framework also has structural limitations. It evaluates discrepancies at the level of individual assertions, which may not capture all forms of harmful content. Governing references may be incomplete, disputed, or unavailable, particularly in tasks involving subjective judgment or creative expression. Retrieval, packing, generation, and intervention may be difficult to distinguish in tightly integrated systems. Authorized project state may likewise be difficult to define and track in informal or multi-party workflows.

Despite these limitations, the framework offers a structured method for evaluating errors in delivered AI responses. By separating governing-reference selection, discrepancy classification, causal attribution, consequence assessment, and governance response, it helps reduce conflation of distinct evaluative questions. By distinguishing representational assertions from task requirements, it clarifies the relationship between hallucination and task-obligation fidelity error. By analyzing retrieval-augmented generation through its effect on E_t and C_t , it integrates retrieval into the production model without altering the authority-first structure. By incorporating time, project state, and possible propagation, it extends evaluation across continuing workflows.

The result is an authority-first evaluation sequence: identify what governs a claim, and what evidence is needed, before classifying a discrepancy, tracing its cause, or deciding on a governance response. The sequence is intended to support clearer diagnosis, more targeted governance actions, and more reliable deployment of generative AI systems. Whether it produces more consistent evaluations or better control choices remains an empirical question.

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